

LODZ, Poland

WMO: 124650

Lat: 51.723N

Lon: 19.399E

Elev: 623

StdP: 14.37

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	10.8	0.6	5.8	6.9	5.6	7.5	12.5	25.0	39.4	22.9	37.8	5.2	100	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	86.5	67.7	82.9	66.2	80.0	64.9	70.0	81.4	68.4	79.1	66.8	76.3	8.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	99.7	73.7	64.8	93.8	71.9	63.3	88.9	70.6	34.3	81.2	33.0	79.1	31.7	76.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.8	15.7	DB	-0.6	92.6	7.7	3.2	-6.1	94.9	-10.6	96.7	-14.9	98.5	-20.5	100.9	(4)
(5)			WB	-1.2	72.9	7.5	1.7	-6.5	74.2	-10.9	75.1	-15.1	76.1	-20.5	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	28.7	31.3	37.5	48.0	56.6	62.3	66.3	65.6	56.9	47.9	39.3	31.9	(6)
(7)		DBStd	15.24	9.63	8.90	7.95	7.86	7.14	6.42	6.16	5.87	6.27	7.40	7.48	8.81	(7)
(8)		HDD50	2749	660	523	391	126	20	0	0	0	11	130	327	561	(8)
(9)		HDD65	6537	1125	943	852	511	275	126	59	65	253	530	772	1026	(9)
(10)		CDD50	1941	0	0	4	67	223	369	506	484	217	66	5	0	(10)
(11)		CDD65	254	0	0	0	2	14	45	100	84	9	0	0	0	(11)
(12)		CDH74	2526	0	0	0	29	180	490	969	752	105	1	0	0	(12)
(13)		CDH80	697	0	0	0	3	28	132	300	213	21	0	0	0	(13)
(14)	Wind	WSAvg	7.1	8.4	8.0	8.1	7.3	6.6	6.5	6.3	5.7	6.2	6.9	7.5	8.2	(14)
(15)	Precipitation	PrecAvg	22.8	1.4	1.3	1.5	1.4	2.5	2.6	3.3	2.2	1.9	1.5	1.7	1.5	(15)
(16)		PrecMax	29.6	2.9	2.9	3.1	3.5	6.2	6.3	9.8	5.1	4.4	5.0	3.9	3.5	(16)
(17)		PrecMin	16.0	0.2	0.2	0.6	0.0	0.8	0.5	0.9	0.7	0.5	0.2	0.0	0.3	(17)
(18)		PrecStd	3.7	0.6	0.6	0.6	0.7	1.3	1.6	2.1	1.0	0.9	1.1	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	53.0	63.4	78.1	83.4	89.4	91.9	91.2	82.8	71.6	59.1	52.0	(19)
(20)			MCWB	46.2	46.2	50.9	59.7	64.2	69.2	68.1	68.3	64.9	60.9	52.6	48.5	(20)
(21)		2%	DB	46.2	49.2	58.0	71.5	78.9	83.8	87.0	85.4	76.6	67.1	55.5	48.2	(21)
(22)			MCWB	43.6	44.8	48.0	56.4	62.6	66.6	67.7	67.8	62.4	58.5	51.5	45.5	(22)
(23)		5%	DB	43.2	46.1	53.7	66.3	74.9	79.4	83.1	81.5	72.7	62.9	52.2	45.4	(23)
(24)			MCWB	41.0	42.9	46.2	53.5	61.0	64.3	66.5	66.1	61.0	56.4	49.2	43.5	(24)
(25)		10%	DB	40.8	42.9	49.7	62.3	70.8	75.6	79.4	77.9	68.9	59.1	49.8	42.8	(25)
(26)			MCWB	39.0	40.2	44.4	51.4	58.6	62.6	65.2	64.5	59.4	54.6	47.3	41.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	47.9	52.2	61.2	66.6	71.1	72.5	72.0	66.9	62.6	54.7	49.4	(27)
(28)			MCD B	48.9	50.5	61.1	74.9	78.9	83.7	84.1	82.9	78.1	69.0	57.8	51.3	(28)
(29)		2%	WB	43.6	45.6	49.4	57.7	64.3	68.4	70.3	69.6	64.3	59.5	52.2	45.9	(29)
(30)			MCD B	45.5	48.4	55.6	69.4	75.2	79.9	81.4	80.9	72.9	65.2	54.9	47.6	(30)
(31)		5%	WB	41.3	42.9	47.1	55.0	62.3	66.3	68.7	67.9	62.4	57.3	49.8	43.5	(31)
(32)			MCD B	43.1	45.6	52.4	63.9	73.0	76.0	78.8	78.6	70.3	62.1	52.0	45.1	(32)
(33)		10%	WB	38.8	40.2	44.6	52.5	60.1	64.3	67.0	66.2	60.4	55.0	47.4	41.1	(33)
(34)			MCD B	40.4	42.6	49.0	60.1	68.7	73.4	75.9	75.1	67.1	58.9	49.4	42.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	10.6	13.9	17.9	18.9	18.8	18.6	19.1	17.0	14.0	9.5	8.1	(35)
(36)			MCD BR	9.6	14.2	21.4	25.7	25.2	25.5	25.1	24.6	23.9	20.1	14.4	9.7	(36)
(37)			MCWBR	8.0	10.4	13.2	13.1	11.7	11.4	9.5	10.3	11.7	12.4	10.5	8.0	(37)
(38)		5% WB	MCD BR	9.4	13.1	19.2	22.4	22.6	21.9	20.5	21.6	20.7	18.5	13.2	9.3	(38)
(39)			MCWBR	8.2	10.3	12.7	12.2	11.4	11.0	9.2	10.0	11.1	11.9	10.3	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.343	0.379	0.431	0.422	0.417	0.447	0.435	0.398	0.369	0.350	0.313	(40)	
(41)		taud	2.260	2.260	2.216	2.127	2.206	2.253	2.182	2.219	2.289	2.341	2.319	2.301	(41)	
(42)		Ebn at Noon	213	242	255	254	264	266	255	251	246	230	199	195	(42)	
(43)		Edh at Noon	22	28	36	44	43	41	43	40	33	26	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	229	449	838	1315	1642	1718	1602	1431	1025	576	260	174	(44)	
(45)		RadStd	43	80	102	174	172	164	190	120	142	88	34	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.92	N/A	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-295	+140	+56

Nomenclature: See separate page